# CMS | Next-year inpatient claims ML

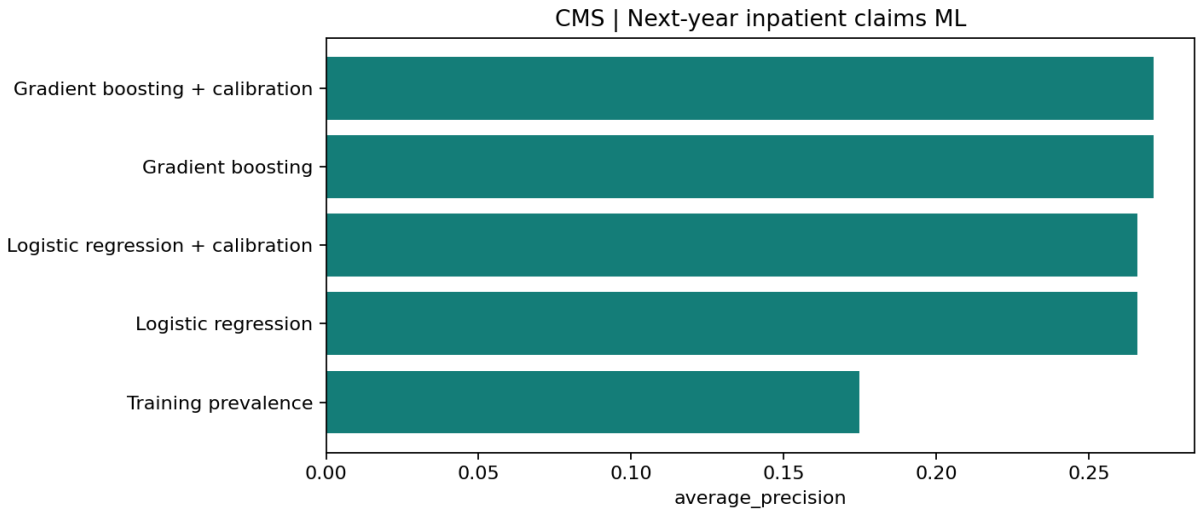
## Scope

CMS DE-SynPUF Sample 1, synthetic data only. 73,757 eligible beneficiaries. Baseline features end 2008-12-31; target is at least one accepted claim ending in 2009. Full 2009 Part A, no HMO and survival through 2009 are required to interpret absence as zero.

## Validation

Fixed seed, stratified beneficiary split: 60% training, 20% probability calibration, 20% locked test. Fixed hyperparameters; median imputation and scaling fitted on training only. Platt calibration uses calibration people only. No test-driven model selection. This is a held-out-person evaluation with forward-looking labels, not validation in a later calendar cohort.

| model | average_precision |
|---|---|
| Training prevalence | 0.1749 |
| Logistic regression | 0.2660 |
| Logistic regression + calibration | 0.2660 |
| Gradient boosting | 0.2711 |
| Gradient boosting + calibration | 0.2711 |




## Interpretation and limits

Synthetic relationships do not demonstrate performance on real patients or clinical usefulness. Eligibility depends on later enrollment and survival, so results apply to that retrospective complete-follow-up population. Labels include carry-in stays ending in 2009, not incident admissions or readmissions. Aggregate held-out errors have no external validation or confidence intervals; subgroup scores do not establish fairness.